

2001

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 13, 17 and 21

Total Marks – 100

Section I Pages 2–24

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–31

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

1 Ethene may be converted into poly(ethene).

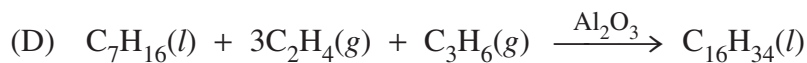
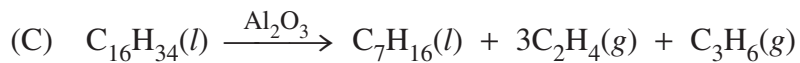
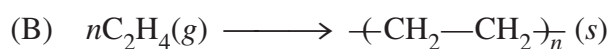
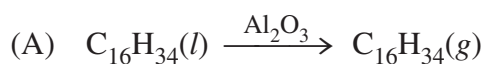
What type of reaction is this?

- (A) Condensation
- (B) Hydrolysis
- (C) Oxidation/reduction
- (D) Polymerisation

2 Which of the following is a major component of biomass?

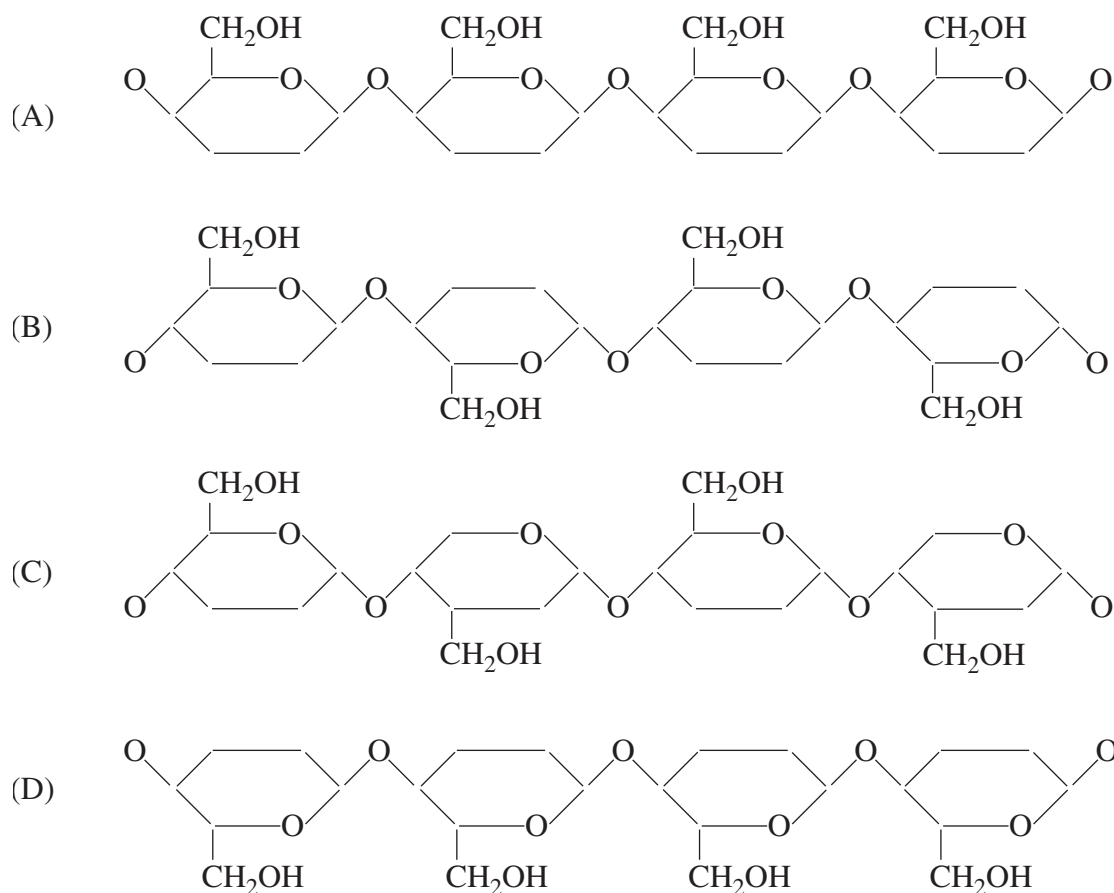
- (A) Cellulose
- (B) Ethanol
- (C) Natural gas
- (D) Oil

3 Which equation best represents catalytic cracking of a petroleum fraction?



- 4 Cellulose is a linear polymer which is a basic structural component of plant cell walls.

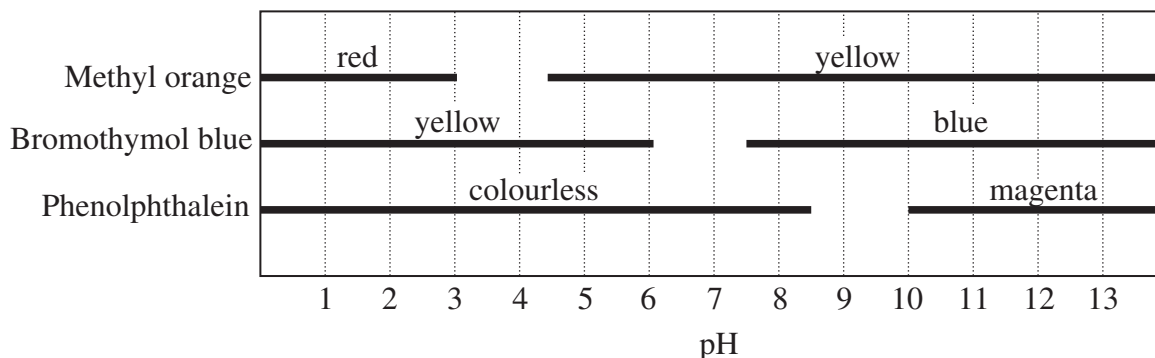
Which is the correct representation of part of a cellulose polymer?



- 5 The pH of unpolluted rainwater is about 6.0. Which substance contributes most to this?

- (A) CO_2
(B) N_2
(C) NO_2
(D) O_3

- 6 The graph shows the colour ranges of the acid–base indicators methyl orange, bromothymol blue and phenolphthalein.



A solution is yellow in methyl orange, blue in bromothymol blue and colourless in phenolphthalein.

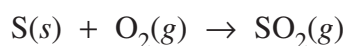
What is the pH range of the solution?

- (A) 4.5 to 6.0
 - (B) 6.0 to 7.5
 - (C) 7.5 to 8.5
 - (D) 8.5 to 10.0
- 7 A group of students produced a red solution by boiling red cabbage leaves in water. When dilute sodium hydroxide was added to the solution, it turned purple. When dilute hydrochloric acid was added to the red solution, no colour change occurred.

Which of these substances, when added, is most likely to cause the red solution to change colour?

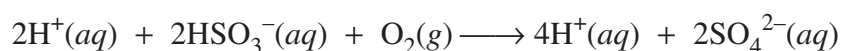
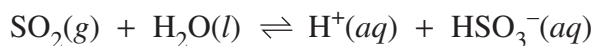
- (A) Cleaning solution containing ammonia
- (B) Concentrated hydrochloric acid
- (C) Orange juice
- (D) Vinegar

- 8 The burning of sulfur can be described by the following equation:



What volume of sulfur dioxide gas will be released at 25°C and 101.3 kPa when 8.00 g of sulfur is burnt?

- (A) 3.06 L
 - (B) 6.12 L
 - (C) 12.24 L
 - (D) 24.47 L
- 9 An understanding of Le Chatelier's principle is important in the chemical industry. Which prediction can be made using this principle?
- (A) The identity of products of a chemical reaction
 - (B) The effect of changes in temperature on the rates of reactions
 - (C) The effect of catalysts on the position of equilibrium reactions
 - (D) The effect of changes in the concentration of chemical substances in equilibrium
- 10 The following equations describe some reactions in the formation of acid rain:



What would occur if some solid sodium sulfate (Na_2SO_4) were added to a sample of acid rain?

- (A) The amount of $\text{SO}_2(g)$ would increase and the acidity of the solution would decrease.
- (B) The amount of $\text{SO}_2(g)$ would increase and the acidity of the solution would increase.
- (C) The amount of $\text{SO}_2(g)$ would be unchanged and the acidity of the solution would be unchanged.
- (D) The amount of $\text{SO}_2(g)$ would be unchanged and the acidity of the solution would decrease.

- 11 Why is chlorine used to treat local water supplies?
- (A) To make water suitable for swimming
 (B) To kill micro-organisms living in the water
 (C) To promote sedimentation of finely suspended solids
 (D) To precipitate heavy metal ions such as lead and mercury
- 12 The atomic absorption spectrophotometer was developed by Sir Alan Walsh and his team at CSIRO in the 1950s. Its development was one of the most significant in Australian chemical technology. What did it provide?
- (A) A rapid method to monitor chemical pollutants in water supplies
 (B) The first method for determining the concentrations of metal ions in water supplies
 (C) A method for determining the concentrations of hydrocarbons at very low concentrations
 (D) A method for determining the concentrations of metal ions at very low concentrations
- 13 Four students analysed a sample of fertiliser to determine its percentage of sulfate.
- Each student:
- weighed an amount of fertiliser;
 - dissolved this amount in 100 mL of water;
 - added aqueous barium nitrate;
 - filtered, dried and weighed the barium sulfate precipitate.

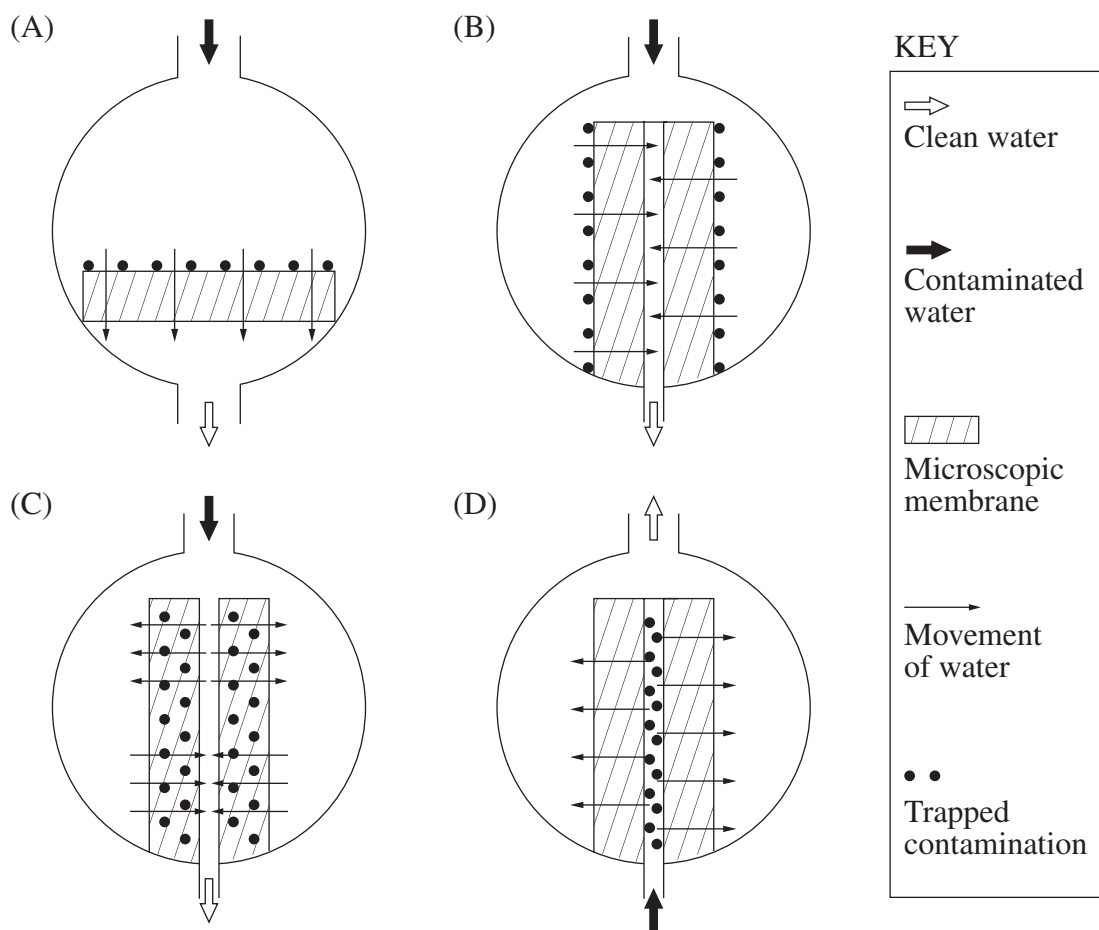
Their results and calculations are shown in the table.

<i>Student</i>	<i>Mass of fertiliser used (g)</i>	<i>Mass of BaSO₄ weighed (g)</i>	<i>Percentage of sulfate in fertiliser (%)</i>
<i>A</i>	11.6	19.5	69.2
<i>B</i>	10.4	16.9	66.9
<i>C</i>	10.268	22.612	90.6
<i>D</i>	11.1	18.2	67.5

The percentage of sulfate calculated by Student *C* was significantly higher than that of the other students. Which is the most likely reason for this?

- (A) Student *C* did not dry the sample for long enough.
 (B) Student *C* added more Ba(NO₃)₂ solution than the other students.
 (C) Student *C* used a balance capable of measuring weight to more decimal places.
 (D) Student *C* waited longer than the other students for the Ba(NO₃)₂ to react completely with the sulfate.

- 14 Which diagram represents the most effective design for a microscopic membrane filter to purify contaminated water?



- 15 Four students were asked to test a solution for the presence of a cation by using various anions. The students obtained these results:

<i>Student</i>	<i>Chloride</i>	<i>Sulfate</i>	<i>Carbonate</i>
<i>A</i>	no precipitate	no precipitate	precipitate
<i>B</i>	precipitate	precipitate	no precipitate
<i>C</i>	precipitate	precipitate	precipitate
<i>D</i>	no precipitate	precipitate	no precipitate

Each student concluded that Pb^{2+} was present.

Which student had results consistent with this conclusion?

- (A) *A*
 (B) *B*
 (C) *C*
 (D) *D*

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Centre Number

Section I (continued)

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Part B – 60 marks

Student Number

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (3 marks)

Radioisotopes are used in industry, medicine and chemical analysis. For ONE of these fields, relate the use of a named radioisotope to its properties.

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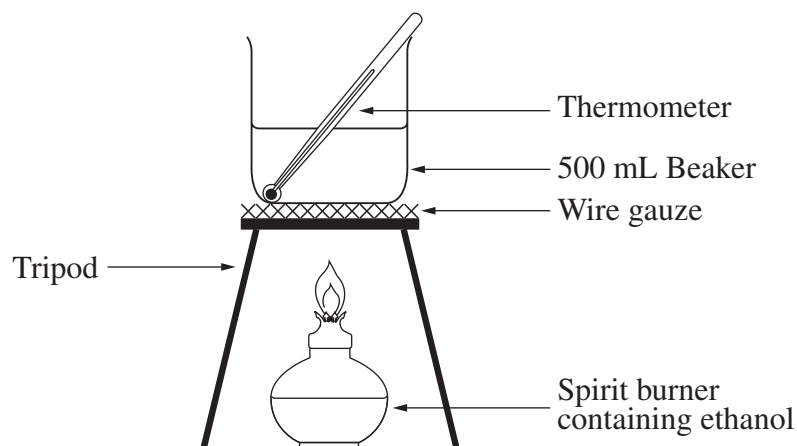
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Question 17 (6 marks)

Students were asked to perform a first-hand investigation to determine the molar heat of combustion of ethanol.

The following extract is from the practical report of one student.

Apparatus used:



Lab data:

Mass of water	=	250.0 g
Initial mass of burner	=	221.4 g
Final mass of burner	=	219.1 g
Initial temperature of water	=	19.0°C
Final temperature of water	=	59.0°C

- (a) After completing the calculations correctly, the student found that the answer did not agree with the value found in data books. Suggest ONE reason for this.

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- (b) Propose TWO adjustments that could be made to the apparatus or experimental method to improve the accuracy of the results.

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Question 17 continues on page 11

Question 17 (continued)

- (c) Calculate the molar heat of combustion of ethanol, using the student's data. **3**

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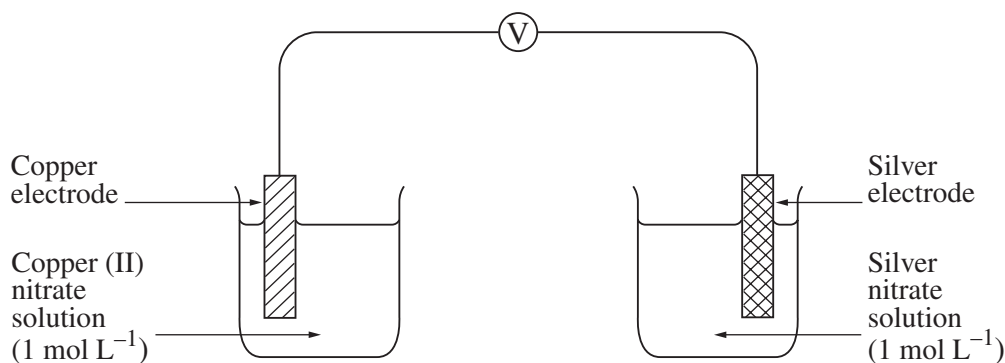
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End of Question 17

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Question 18 (6 marks)

A galvanic cell was made by connecting two half-cells. One half-cell was made by putting a copper electrode in a copper (II) nitrate solution. The other half-cell was made by putting a silver electrode in a silver nitrate solution. The electrodes were connected to a voltmeter as shown in the diagram.



- (a) Complete the above diagram by drawing a salt bridge. 1
- (b) Using the *standard potentials* table in the data sheet, calculate the theoretical voltage of this galvanic cell. 2
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- (c) A student removes the voltmeter from the circuit and replaces it with an electrical generator. The generator causes the copper electrode to increase in mass. 3

Explain, using an equation, why the copper electrode will increase in mass.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 19 (7 marks)

Name ONE type of cell, other than the dry cell or lead–acid cell, you have studied. Evaluate it in comparison with either the dry cell or lead–acid cell, in terms of chemistry and the impact on society. Include relevant chemical equations in your answer.

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Question 20 (4 marks)

A 0.1 mol L^{-1} solution of hydrochloric acid has a pH of 1.0, whereas a 0.1 mol L^{-1} solution of citric acid has a pH of 1.6.

- (a) State ONE way in which pH can be measured. **1**

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- (b) Explain why the two solutions have different pH values. **3**

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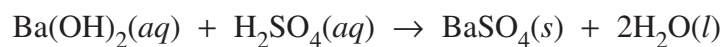
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Question 21 (4 marks)

Barium hydroxide and sulfuric acid react according to the following equation:

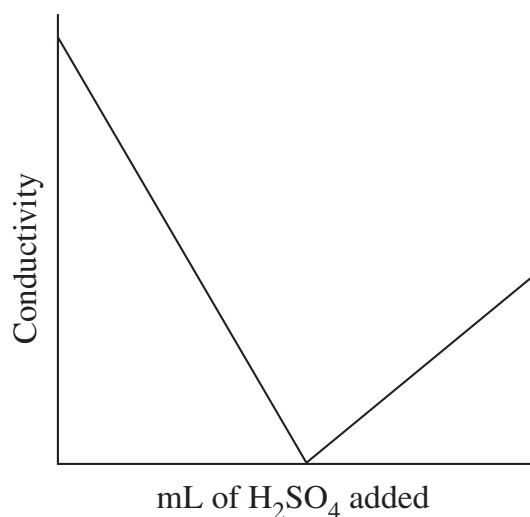


- (a) Name this type of chemical reaction.

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- (b) A 20 mL sample of barium hydroxide was titrated with 0.12 mol L^{-1} sulfuric acid. The conductivity of the solution was measured throughout the titration and the results graphed, as shown.

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Explain the changes in conductivity shown by the graph.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 22 (6 marks)

Justify the procedure you used to prepare an ester in a school laboratory. Include relevant chemical equations in your answer.

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Question 23 (4 marks)

A household cleaning agent contains a weak base of general formula NaX. 1.00 g of this compound was dissolved in 100.0 mL of water. A 20.0 mL sample of the solution was titrated with $0.1000 \text{ mol L}^{-1}$ hydrochloric acid and required 24.4 mL of the acid for neutralisation.

- (a) What is the Brønsted–Lowry definition of a base? **1**

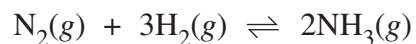
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- (b) What is the molar mass of this base? **3**

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Question 24 (6 marks)

In the early twentieth century, Fritz Haber developed a method for producing ammonia, as shown by the equation:



- (a) Ammonia is used as a cleaning agent. State ONE other use of ammonia. **1**

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- (b) Explain the effect of liquefying the ammonia on the yield of the reaction. **2**

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- (c) Explain why it is essential to monitor the temperature and pressure inside the reaction vessel. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 25** (6 marks)

Explain the need for monitoring the products of a chemical reaction such as combustion.

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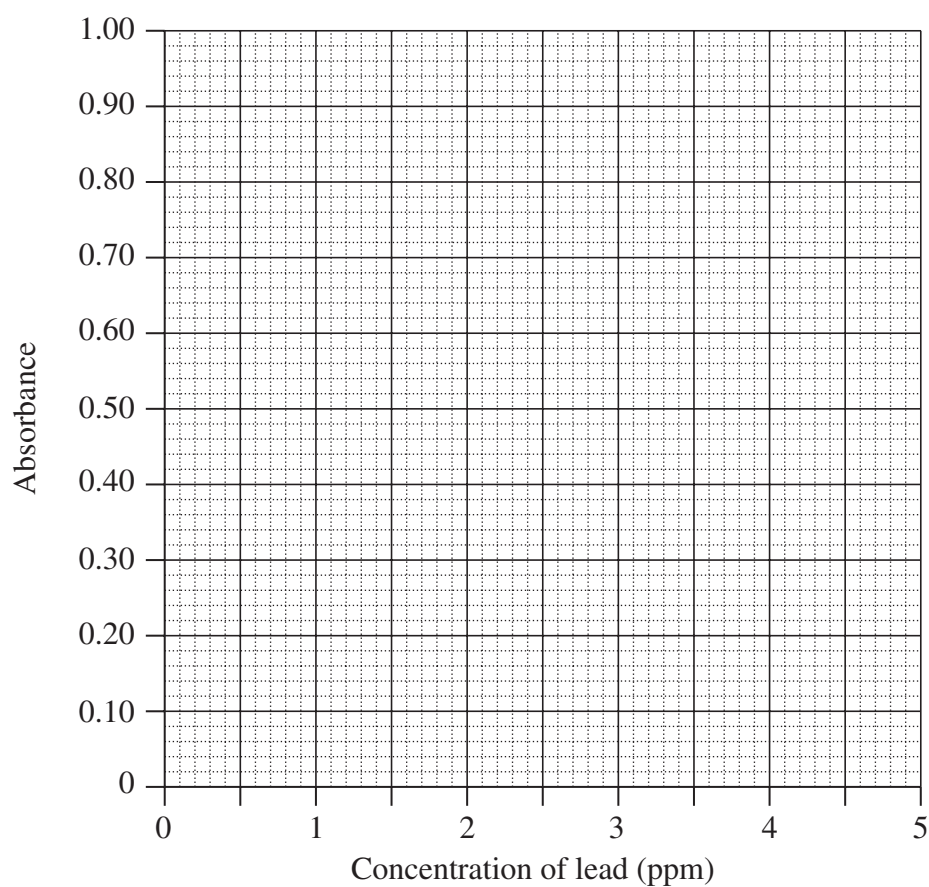
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Question 26 (4 marks)

A university student decided to measure the concentration of lead (Pb) in the soil around his home. He prepared five standard lead solutions of known concentration. The absorbance of these solutions was measured. These results are shown in the table.

<i>Concentration of lead standard (ppm)</i>	<i>Absorbance</i>
0	0.00
1	0.15
2	0.31
3	0.44
4	0.59
5	0.75

- (a) Draw a line graph of these data.

1

Question 26 continues on page 23

Question 26 (continued)

- (b) The student prepared solutions from four different soil samples around his home. These solutions were also analysed using the same method. The results are shown in the table. 1

<i>Solutions made from soil samples</i>	
<i>Area sampled</i>	<i>Absorbance</i>
Front garden bed	0.19
Back garden bed	0.09
Mail box	0.22
Back fence	0.11

Determine the highest concentration of lead in the soil around the home.

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- (c) State an hypothesis to account for the variation in lead concentration around the student's home. 2

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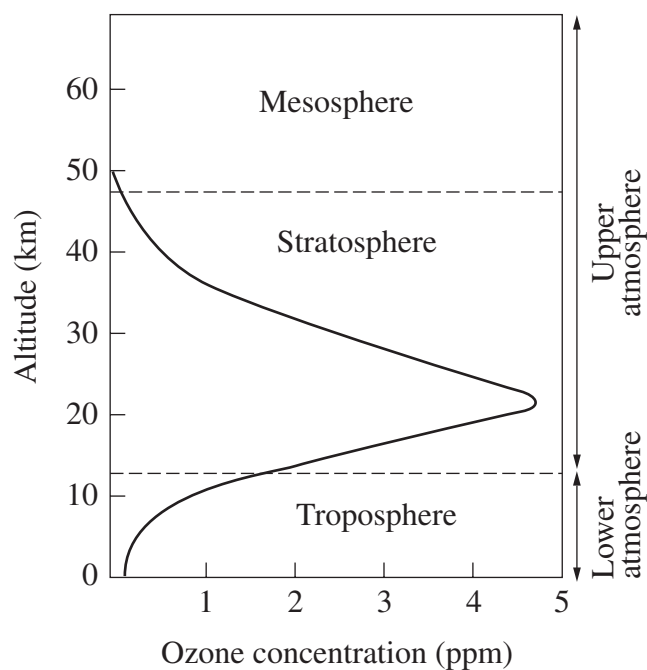
End of Question 26

Please turn over

Question 27 (4 marks)

Oxygen exists in the atmosphere as the allotropes oxygen and ozone. The graph shows a typical change in ozone concentration with changing altitude.

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Compare the environmental effects of the presence of ozone in the upper and lower atmosphere.

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

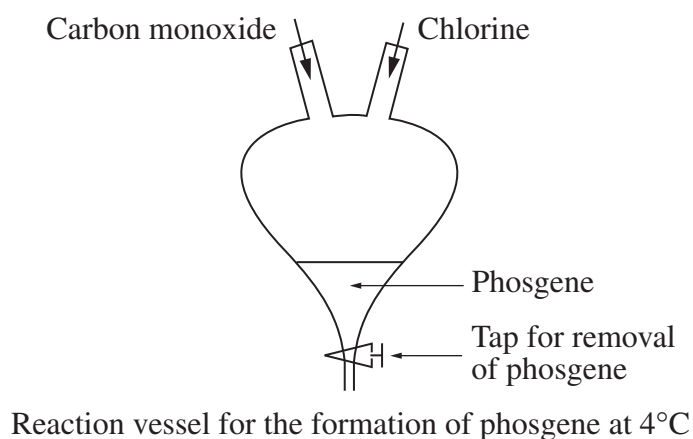
Show all relevant working in questions involving calculations.

	Pages
Question 28 Industrial Chemistry	26
Question 29 Shipwrecks and Salvage	27
Question 30 The Biochemistry of Movement	28
Question 31 The Chemistry of Art	29–30
Question 32 Forensic Chemistry	31

Question 28 — Industrial Chemistry (25 marks)

Electrolysis is an important industrial process.

- (a) (i) Define *electrolysis*. 1
- (ii) Compare the reaction products from the electrolysis of molten sodium chloride and concentrated aqueous sodium chloride. 2
- (b) Carbonyl chloride, COCl_2 , is a colourless, poisonous gas that is also known as phosgene. It is needed for the production of insecticides, polyurethane plastics and polycarbonate. It is produced from the exothermic equilibrium reaction of carbon monoxide gas and chlorine gas. When the reaction vessel is cooled below 8°C the phosgene is a liquid.



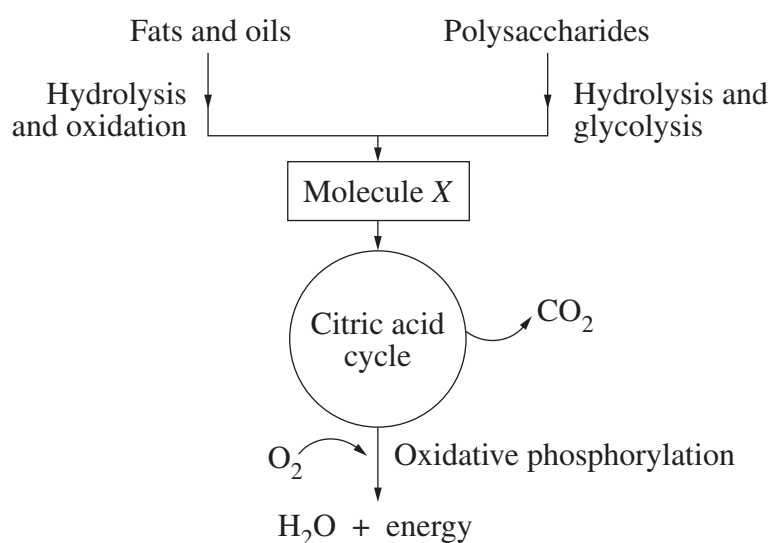
- (i) Write a balanced equation for the formation of phosgene. 2
- (ii) Explain how industry could maximise the production of phosgene. 2
- (c) Explain why sulfuric acid is an important industrial chemical. Include balanced chemical equations in your answer. 5
- (d) (i) Name the chemical process used to make soap. 1
- (ii) Outline the procedure for making soap in the school laboratory. 2
- (iii) Describe a safety risk associated with the procedure outlined in part (ii), and suggest a safe work practice to minimise the risk. 3
- (e) Evaluate how environmental issues are addressed in the Solvay process. 7

Question 29 — Shipwrecks and Salvage (25 marks)

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|-----|-------|---|----------|
| (a) | (i) | Identify the main metal used to construct ships. | 1 |
| | (ii) | Although aluminium is a very reactive metal, with a very low reduction potential, it is used in many structures exposed to oxidising conditions. Explain why aluminium can be used in this way. | 2 |
| (b) | (i) | Give an example of a metal commonly used as a sacrificial anode. | 1 |
| | (ii) | Explain why sacrificial anodes are added to metal-hulled ships. | 3 |
| (c) | | Describe the effect of adding other elements to iron on the properties and uses of steels. | 5 |
| (d) | (i) | Define <i>corrosion</i> . | 1 |
| | (ii) | Outline a procedure that could be used to compare the corrosion rates of different metals or alloys in the school laboratory. | 2 |
| | (iii) | Describe ways in which accuracy and reliability could be improved in the procedure described in part (ii). | 3 |
| (e) | | For ONE specific metal, evaluate the steps that can be used to clean, stabilise and preserve artefacts recovered from shipwrecks. | 7 |

Question 30 — The Biochemistry of Movement (25 marks)

- (a) (i) Name the molecule that stores energy for nearly all metabolic processes. **1**
- (ii) Explain how the biologically important part of the molecule in part (i) provides energy for cellular metabolism. **2**
- (b) Energy for our bodies to function is provided by the oxidation of fuels. Fats and glucose are used as fuels. A summary of the oxidation of fats and glucose is shown.

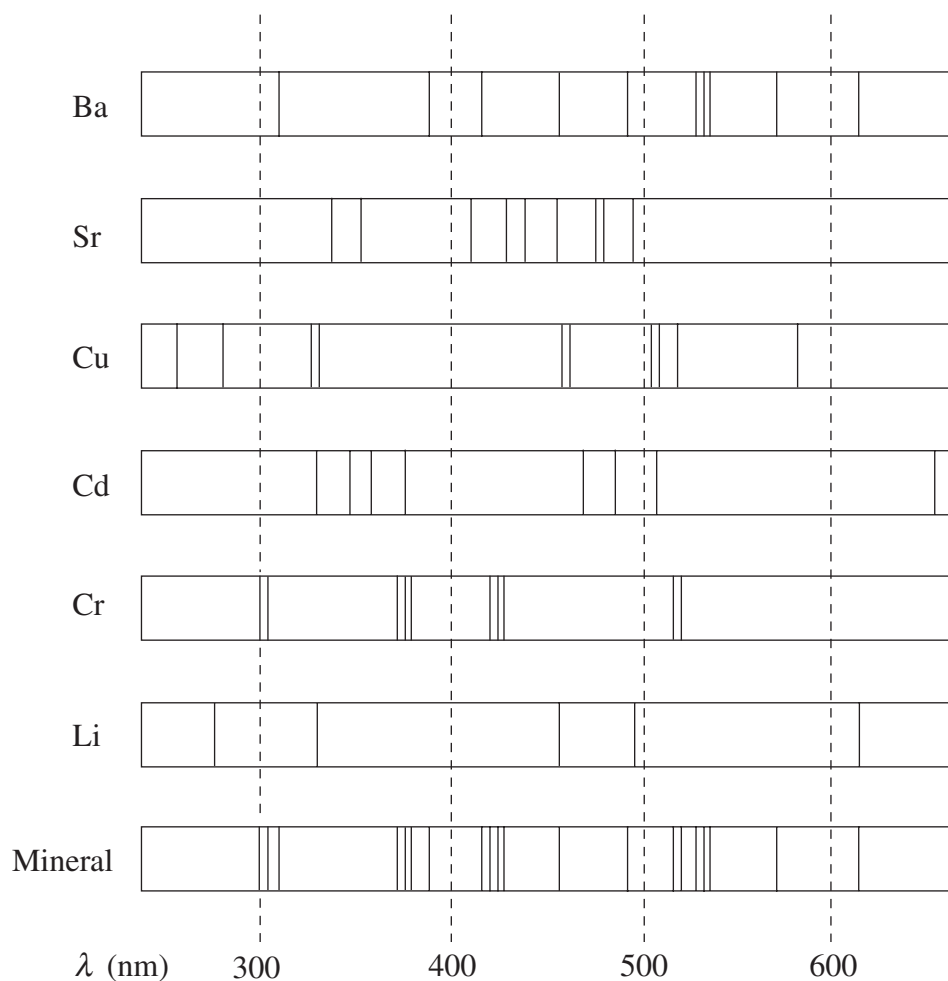


- (i) Identify molecule X and state its function. **2**
- (ii) Analyse the role of oxidative phosphorylation in energy production. **2**
- (c) Discuss the use of models in developing an understanding of enzyme function. **5**
- (d) (i) Define *viscosity*. **1**
- (ii) Outline a procedure that could be used to compare the viscosity of pure glycerol and a glycerol solution. **2**
- (iii) Describe ways in which accuracy and reliability could be improved in the procedure described in part (ii). **3**
- (e) The energy requirements for different types of skeletal muscle are met by the interaction of separate energy output systems. **7**

Analyse the role and interaction of the energy output systems used by skeletal muscle.

Question 31 — The Chemistry of Art (25 marks)

- (a) (i) Identify the mineral source of a pigment. 1
- (ii) Describe the use of a named separation process to obtain a pigment. 2
- (b) A sample of a mineral used as a pigment was subjected to spectroscopic analysis. The spectrum is shown.



- (i) Identify the elements in the mineral sample. 2
- (ii) Explain how a line spectrum is produced. 2

Question 31 continues on page 30

	Marks
Question 31 (continued)	
(c) Describe the Bohr model of the atom, and identify ONE merit and ONE limitation of the model.	5
(d) (i) Name a transition element.	1
(ii) Outline a first-hand investigation to demonstrate the colour changes of a named transition element as it changes in oxidation state.	2
(iii) Describe a safety risk associated with the procedure outlined in part (ii), and suggest a safe work practice to minimise the risk.	3
(e) With reference to TWO pigments, explain how the metallic components produce colour.	7

End of Question 31

Question 32 — Forensic Chemistry (25 marks)

(a) (i) Define *organic compounds*. **1**

(ii) For ONE class of organic compound, describe a chemical test that identifies this class. **2**

(b) The table shows fatty acid composition of some common oils and fats.

<i>Fatty acids present (% by weight)</i>					
Fat or oil hydrolysed	Lauric	Palmitic	Stearic	Oleic	Linoleic
Butter	2–3	23–26	10–13	30–40	4–5
Lard	< 1	28–30	12–18	41–48	6–7
Tallow	< 1	24–32	14–32	35–38	2–4
Coconut	45–51	4–10	1–5	2–10	0–2

An oily sample was hydrolysed and the fatty acids analysed.

<i>Fatty acids present (% by weight)</i>					
	Lauric	Palmitic	Stearic	Oleic	Linoleic
Oil sample	< 1	29	28	36	4

(i) Which fat or oil has been identified? **1**

(ii) Explain the solubility in water of fatty acids, in terms of their structure. **3**

(c) Assess the usefulness of mass spectrometry in providing forensic evidence. **5**

(d) (i) Name ONE technique used by forensic chemists to separate a mixture of organic compounds. **1**

(ii) In your study of Forensic Chemistry, you performed a first-hand investigation to separate a mixture of organic materials. Outline the procedure. **2**

(iii) Describe ways in which accuracy and reliability could be improved in the procedure described in part (ii). **3**

(e) Discuss the uses of DNA analysis in forensic chemistry. **7**

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DATA SHEET

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 101.3 kPa (1.00 atm) and	
at 273 K (0°C)	22.41 L
at 298 K (25°C)	24.47 L
Ionisation constant for water at 298 K (25°C), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS															
KEY															
Atomic Number															
Symbol of element															
Atomic Weight															
Name of element															
1 H 1.008 Hydrogen	4 Be 9.012 Beryllium	79 Au 197.0 Gold	8 O 16.00 Oxygen	9 F 19.00 Fluorine	2 He 4.003 Helium										
3 Li 6.941 Lithium	12 Mg 24.31 Magnesium	7 N 14.01 Nitrogen	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	10 Ne 20.18 Neon										
11 Na 22.99 Sodium	20 Ca 40.08 Calcium	15 P 30.97 Phosphorus	34 Se 78.96 Selenium	35 Br 79.90 Bromine	18 Ar 39.95 Argon										
19 K 39.10 Potassium	38 Sr 87.62 Strontium	33 As 74.92 Arsenic	52 Te 127.6 Tellurium	53 I 126.9 Iodine	36 Kr 83.80 Krypton										
37 Rb 85.47 Rubidium	88.91 Yttrium	51 Sb 121.8 Antimony	84 Po [210.0] Polonium	85 At [210.0] Astatine	54 Xe 131.3 Xenon										
55 Cs 132.9 Caesium	57–71 Lanthanides	83 Bi 209.0 Bismuth	82 Pb 207.2 Lead	86 Rn [222.0] Radon	8 O 16.00 Oxygen										
87 Fr [223.0] Francium	89–103 Actinides	115 — Ununpentium	114 Uuq — Ununquadium	117 — Ununseptium	10 Ne 20.18 Neon										

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.

2001 HSC Notes from
the Examination Centre
Chemistry

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2001 HSC NOTES FROM THE EXAMINATION CENTRE CHEMISTRY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It provides comments with regard to responses to the 2001 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2001 Higher School Certificate Examination, the Marking Guidelines and other support documents, which have been developed by the Board of Studies to assist in the teaching and learning of Chemistry.

In 2001, 9017 candidates presented for the examination in Chemistry.

General Comments

Most candidates were successful in responding to questions that involved the provision of specific knowledge, rather than questions requiring evaluation, justification or explanation.

Some candidates had difficulty using the language of chemistry ie chemical equations. Equations should be used to help chemical communication. This can include answers to questions that do not specifically include the direction 'include relevant chemical equations in your answer'.

Many candidates had difficulty setting out calculations in a logical fashion so that they could be easily understood. Others had difficulty restricting their answers to the ruled space provided on the examination paper. It should be noted that point-form presentation, and the use of diagrams and/or tables, when appropriate, enables candidates to communicate their knowledge and understanding in a logical and coherent manner.

It was obvious from some responses that some first-hand investigations had not been done by candidates. In the options, questions requiring candidates to distinguish between accuracy and reliability were poorly answered.

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills, developed through the study of discrete sections, should accumulate to a more comprehensive understanding than may be described in each section separately.

Section I – Core**Part A – Multiple Choice**

Questions	Correct Response
1	D
2	A
3	C
4	B
5	A
6	C
7	A
8	B

Questions	Correct Response
9	D
10	C
11	B
12	D
13	A
14	B
15	C

Part B**Question 16**

In general, this question was answered quite well, with relatively few candidates scoring 0 or 1 marks out of three. Many candidates had difficulty recalling more than one property. Candidates who did so, usually only related one property to the radioisotope's use. A significant number of candidates included nuclear equations unnecessarily.

Question 17

- (a) A significant proportion of candidates gave very general responses such as 'inaccurate measuring' or 'human error', which did not provide sufficient detail to be awarded the mark.
- (b) Some candidates suggested replacing the beaker with a styrofoam cup. This may demonstrate knowledge of insulating properties but little experience with practical first-hand investigation.
- (c) Most candidates successfully attempted the calculation although many used 4.18 or 4.2 rather than the value $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$ provided in the data sheet.

Question 18

- (a) Almost all candidates drew an identifiable salt bridge although many lacked appropriate detail.
- (b) Even though copper(II) was specified, a sizeable number of candidates used a half equation involving copper(I). Candidates who used $E_{\text{total}} = E_{\text{red}} - E_{\text{ox}}$ rather than manipulating half equations were more prone to errors in their calculations.
- (c) Many candidates showed a good understanding of the concepts but failed to identify that the applied external voltage was responsible for reversing the reaction.

Question 19

Many candidates need to better organise their response to questions requiring a sustained response. Candidates should spend at least a few minutes clarifying what the question requires, and planning how they will structure their response in the space provided. Candidates demonstrated a better knowledge of electrode materials than of the electrolytes used.

Question 20

- (a) Most candidates answered this part correctly.
- (a) Many candidates did not explain why the two solutions had different pH values.

Question 21

- (a) This question part was answered well.
- (b) Many candidates merely described the changes apparent in the graph rather than explaining the changes.

Question 22

Many candidates had difficulty identifying appropriate justifications for the procedure steps outlined.

The existence of dotted lines in the answer booklet does not preclude drawing a diagram to describe a procedural step. Candidates who drew a small, well-labelled diagram could describe the step more succinctly.

Many candidates were confused over the use of concentrated sulfuric acid. A surprising number used calcium carbonate rather than porous ceramic material as boiling chips.

Question 23

- (a) This question part was well done.
- (b) The setting out of answers was often quite poor, with a lack of logical reasoning. Common errors were: not providing a unit for the molar mass, and not allowing for the dilution factor (20 mL of 100 mL prepared solution titrated). Many candidates calculated the molar mass of the X rather than the base NaX used to prepare the solution.

Question 24

- (a) Many responses to this part were too vague.
- (b) Rather than providing an explanation, as required, many candidates described what happened. Many thought the liquefying occurred in the actual reaction chamber by cooling it, while others thought that the liquefying was caused by increasing pressure within the reaction chamber. The removal of ammonia on liquefaction from the equilibrium equation was missed by many, and was not related to an increase in yield.

- (c) The compromise conditions were often stated, but students then had difficulty explaining the effects of temperature and pressure on both yield and rate.

Question 25

Most candidates' responses focused on the monitoring of a combustion reaction. The full range of marks was also awarded to responses based upon other suitable reactions such as ammonia synthesis.

The better responses demonstrated extensive knowledge of possible reaction products, either due to identified varying reaction conditions or due to the presence of impurities. They also explained the need to monitor the products to avoid problems.

Question 26

- (a) This part was well done.
- (b) This part was also answered well by most candidates.
- (c) Only a minority of candidates responded with a hypothesis. A small proportion of candidates gained the allocated marks.

Question 27

Some candidates wrote extensively about the diagram in a way that was unrelated to the question asked. The effect of the very different concentrations in the upper and lower atmosphere was insufficiently recognised in candidate responses.

Section II – Options

Question 28 – Industrial Chemistry

- (a)
 - (i) This part was poorly answered by many candidates.
 - (ii) This part, focusing on a 'column three' activity of the syllabus, was very poorly answered.
- (b)
 - (i) This part was well done.
 - (ii) Some candidates wrote excessively — up to three or four pages — in responding to this two-mark question.
- (c) Good responses gave equations and a variety of uses/properties. Poor responses tended not to include equations.
- (d)
 - (i) This part was well answered by a majority of candidates.

- (ii) This question part was generally well answered; however, many candidates failed to identify any of the equipment being used during the procedure outlined.
- (iii) Generally, this part was well answered. A common error was not clearly identifying the risk associated with the hazard named.
- (e) The better responses demonstrated an extensive knowledge of the chemistry of the Solvay process, and made judgements about the ways in which associated environmental issues may be addressed.

Weaker responses identified reactants and/or products of the Solvay process or stated a problem associated with the process.

Question 29 – Shipwrecks and Salvage

- (a) (i) This part was very well done.
- (ii) This part was well done.
- (b) (i) This part was well done.
- (ii) The concept of a sacrificial anode was well understood but many candidates did not relate this to reduction potentials, required to gain full marks.
- (c) This part was well done by many candidates. Some candidates had a tendency to write long responses that contained little real chemistry.
- (d) (i) Some candidates restricted their answer to the formation of rust.
- (ii) All candidates need to write about ‘what they did’ rather than just give results when asked to outline a procedure.
- (iii) Many candidates had difficulty distinguishing between the meanings of ‘accuracy’ and ‘reliability’.
- (e) Candidates needed to distinguish between the terms ‘clean’, ‘stabilise’ and ‘preserve’. The better responses emphasised chemical processes including complete descriptions, and provided balanced equations for techniques such as acid cleaning and electrolysis.

Question 30 – The Biochemistry of Movement

- (a) (i) Most candidates readily identified ATP but a few confused it with glucose.
- (ii) Candidates answering the first part correctly generally also answered this part correctly.
- (b) (i) This part was well done.
- (ii) Better candidates explained the coupling with oxidation of NADH and FADH₂.

- (c) Most candidates could name the 'lock and key' model but had problems relating the model to substrate specificity and catalytic properties. Many candidates used the terms 'specificity', 'active site' and 'substrate' but could not explain their significance.
- (d)
 - (i) Few candidates could define viscosity, most using analogies such as thickness.
 - (ii) Many candidates described a suitable method but it was questionable as to whether or not they had carried out the procedure in class.
 - (iii) Most candidates could suggest an improvement but very few could distinguish between reliability and accuracy.
- (e) Many candidates analysed structure rather than energy output systems. Most could identify muscle type and related energy system. Candidates who could describe aerobic and anaerobic pathways did not use flow charts or equations.

Question 31 – The Chemistry of Art

- (a)
 - (i) This part was generally well answered.
 - (ii) Many candidates did not name or describe a separation process. Some candidates described a separation process but did not relate it to a pigment.
- (b)
 - (i) This question was well answered by most candidates.
 - (ii) Detailed answers were generally not given, with candidates often simply stating that electrons were excited. Many candidates did not relate energy transitions to frequency or wavelength.
- (c) Many candidates used general terms not specific to Bohr's model. An answer such as 'did not work for elements other than hydrogen' in describing a limitation of Bohr's model lacks sufficient specificity and detail.
- (d)
 - (i) This part was very well done.
 - (ii) This part was very poorly done. Most candidates either left this part unanswered or chose inappropriate practicals eg flame test. Many did not know reactants for the demonstration, often basing answers on the half equations listed on the data sheet, and inappropriate acids.
 - (iii) Most candidates did not relate the risk assessment to their chosen practical. Candidates found it easier to find a safe work practice than a hazard, and only the better candidates could describe a safety risk.
- (e) Many students discussed only one pigment. Very few students related oxidation state to their explanation of colour and only the best were able to explain the transition between the d orbitals in transition metals. Red ochre was often incorrectly described as $\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$, which is yellow.

Question 32 – Forensic Chemistry

- (a) (i) Many candidates had a poor understanding of the structure of organic compounds and incorrectly focused on the ‘natural origin’ of organic compounds. A significant number of candidates knew that carbon was the main element involved but included oxygen in their definition of elements present. This proved a difficult question; many candidates who scored well in other parts did not gain the mark.
- (ii) Candidates who understood what a ‘class’ of organic compound was answered this question well. A wide range of classes was given. Misunderstanding of the term ‘organic’ led to a surprising number of students referring to organic matter in soil in answering (a)(i), (a)(ii) and (d).
- (b) (i) This part was well answered.
- (ii) Most candidates have a basic knowledge of fatty acid structure and solubility and scored some marks on this question. However many candidates confused triglycerides with fatty acids. Few candidates related the increasing length of the hydrocarbon chain to insolubility.
- (c) This part was well answered, with candidates being able to assess the usefulness of mass spectroscopy.
- (d) (i) This part was well answered by the majority of candidates. Some candidates misread the question and referred to identification of the components of a mixture in (i) and (ii), rather than separation techniques.
- (ii) Most candidates outlined a procedure for chromatography. Candidates should, however, be aware that methods obtained from the internet are not necessarily valid. Where candidates described electrophoresis, it was obvious that this technique had not been performed as a first-hand investigation by many candidates.
- (iii) Most candidates could describe some improvements to accuracy/reliability but very few distinguished between these as was needed to gain full marks.
- (e) Many candidates were unsure of how to write a ‘discussion’ text but merely wrote down everything that they knew, without addressing the question and without any coherent structure.

Chemistry

2001 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1	1	9.2.1	H9
2	1	9.2.2	H9
3	1	9.2.1	H9, H10
4	1	9.2.2	H2, H9
5	1	9.3.2	H8
6	1	9.3.1	H13, H14
7	1	9.3.1	H8
8	1	9.3.2	H10, H14
9	1	9.3.2	H1, H2
10	1	9.3.2	H8, H10
11	1	9.4.5	H3, H4
12	1	9.4.3	H1, H3, H4
13	1	9.4.3	H12, H14
14	1	9.4.5	H3, H12, H14
15	1	9.4.3	H8, H14
16	3	9.2.5	H3, H4, H6
17(a)	1	9.2.3	H7, H9, H10, H14
17(b)	2	9.2.3	H12
17(c)	3	9.2.3	H10, H12, H13
18(a)	1	9.2.4	H13
18(b)	2	9.2.4	H6, H8, H10
18(c)	3	9.2.4	H6, H8, H13, H14
19	7	9.2.4	H1, H3, H4, H5, H7, H8, H13
20(a)	1	9.3.3	H13
20(b)	3	9.3.3	H8, H10, H13
21(a)	1	9.3.4	H8
21(b)	3	9.3.4	H2, H8, H10, H13, H14
22	6	9.3.5	H8, H9, H11, H12, H13
23(a)	1	9.3.4	H1, H6
23(b)	3	9.3.4	H6, H10, H13
24(a)	1	9.4.2	H3
24(b)	2	9.4.2	H8, H13
24(c)	3	9.4.2	H4, H8, H10
25	6	9.4.1	H4, H8, H9, H13
26(a)	1	9.3.4	H13
26(b)	1	9.4.3	H12, H14
26(c)	2	9.4.3	H4, H14
27	4	9.4.4	H4, H5, H13, H14
28(a)(i)	1	9.5.4	H7, H13
28(a)(ii)	2	9.5.4	H7
28(b)(I)	2	9.5.2	H10

Question	Marks	Content	Syllabus outcomes
28(b)(ii)	2	9.5.2	H3, H8
28(c)	5	9.5.3	H3, H8, H10, H13
28(d)(i)	1	9.5.5	H9
28(d)(ii)	2	9.5.5	H12
28(d)(iii)	3	9.5.5	H12
28(e)	7	9.5.6	H4, H8, H13, H14
29(a)(i)	1	9.6.2	H6
29(a)(ii)	2	9.6.2	H6, H7
29(b)(i)	1	9.6.4	H8
29(b)(ii)	3	9.6.4	H4, H8
29(c)	5	9.6.2	H3, H8, H13
29(d)(i)	1	9.6.5	H8, H13
29(d)(ii)	2	9.6.5	H12
29(d)(iii)	3	9.6.5	H12
29(e)	7	9.6.7	H4, H8, H13, H14
30(a)(i)	1	9.7.1	H9
30(a)(ii)	2	9.7.1	H9
30(b)(i)	2	9.7.7	H9
30(b)(ii)	2	9.7.10	H4, H9
30(c)	5	9.7.4	H2, H9, H13
30(d)(i)	1	9.7.3	H13
30(d)(ii)	2	9.7.3	H12
30(d)(iii)	3	9.7.3	H12
30(e)	7	9.7.9	H7, H9, H13, H14
31(a)(i)	1	9.8.1	H8
31(a)(ii)	2	9.8.2	H8
31(b)(i)	2	9.8.3	H6, H14
31(b)(ii)	2	9.8.3	H2, H6
31(c)	5	9.8.3	H2, H6, H13
31(d)(i)	1	9.8.5	H6
31(d)(ii)	2	9.8.5	H12
31(d)(iii)	3	9.8.5	H12
31(e)	7	9.8.2	H6, H13, H14
32(a)(i)	1	9.9.1	H9, H13
32(a)(ii)	2	9.9.1	H9
32(b)(i)	1	9.9.3	H14
32(b)(ii)	3	9.9.3	H4, H9
32(c)	5	9.9.6	H2, H3, H6
32(d)(i)	1	9.9.4	H9
32(d)(ii)	2	9.9.4	H12
32(d)(iii)	3	9.9.4	H12
32(e)	7	9.9.5	H1, H3, H9, H13, H14

2001 HSC Chemistry Marking Guidelines

Question 16 (3 marks)

Outcomes assessed: H3, H4, H6

MARKING GUIDELINES

Criteria	Marks
Correctly names a radioisotope and relates at least two properties to its use	3
- Correctly names a radioisotope and relates at least one property to its use OR - Correctly names a radioisotope and describes two properties OR - Correctly names a radioisotope and describes a use	2
- Correctly names a radioisotope OR - Describes a use of a radioisotope	1

Question 17 (a) (1 mark)

Outcomes assessed: H7, H9, H10, H14

MARKING GUIDELINES

Criteria	Marks
A valid reason	1

Question 17 (b) (2 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Two adjustments given which improve accuracy	2
• One adjustment given which improves accuracy	1

Question 17 (c) (3 marks)

Outcomes assessed: H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Calculates ΔH and the moles of ethanol and calculates $\Delta_c H$ for ethanol	3
• Calculates ΔH and the moles of ethanol OR • Correct method for calculating but minor error in calculation of ΔH or moles	2
• Calculates ΔH or moles of ethanol	1

Question 18 (a) (1 mark)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
• Correctly places salt bridge between the beakers and dipping into each solution	1

Question 18 (b) (2 marks)

Outcomes assessed: H6, H8, H10

MARKING GUIDELINES

Criteria	Marks
• Both half equations/values correct and final calculation correct	2
• At least one half equation value correct	1

Question 18 (c) (3 marks)

Outcomes assessed: H6, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Illustrates that copper ions from solution form copper metal: $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu (s)}$ - Explains that an applied voltage changes the direction of electron flow	3
Gives correct half equation $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$ with a statement that the cell reaction has been reversed	2
- Indicates that metallic copper forms from copper ions OR - States the copper electrode is now a cathode, not an anode OR - States the cell is now electrolytic not galvanic	1

Question 19 (7 marks)

Outcomes assessed: H1, H3, H4, H5, H7, H8, H13

MARKING GUIDELINES

Criteria	Marks
- Evaluates both named cell types in terms of chemistry and impact on society - Answer illustrated with selected balanced symbol equations	6–7
- Names one other cell type and describes a number of features of it and one of the given cells in terms of chemistry and impact on society, illustrating answer with at least word equations and/or formulas OR - Names one other cell type, evaluates it and one of the given cells in one area stated in the question, illustrating answer with word equations and/or formulas, if appropriate	4–5
- Names one other type of cell and describes at least one feature of it and the dry cell or lead-acid cell OR - Describes a number of features of either the dry cell or lead-acid cell	2–3
Names one other type of cell	1

Question 20 (a) (1 mark)*Outcomes assessed: H13***MARKING GUIDELINES**

Criteria	Marks
• Any appropriate pH measuring device named	1

Question 20 (b) (3 marks)*Outcomes assessed: H8, H10, H13***MARKING GUIDELINES**

Criteria	Marks
• Explains the relationship between $[H^+]$ and pH • Indicates that pH 1 means higher $[H^+]$ than pH 1.6 • Explains that HCl ionises more than citric acid	3
• Any two parts of the explanation	2
• Any one part of the explanation	1

Question 21 (a) (1 mark)*Outcomes assessed: H8***MARKING GUIDELINES**

Criteria	Marks
• Any technically correct name given	1

Question 21 (b) (3 marks)

Outcomes assessed: H2, H8, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
Explains original conductivity, the equivalence point and subsequent increase using named chemicals from the reaction	3
- Explains two features of the graph in terms of chemicals involved OR - General explanation involving removal and addition of ions	2
- Explains either the decrease, equivalence point or increase conductivity in terms of chemicals involved OR - General overview in terms of ions given	1

Question 22 (6 marks)

Outcomes assessed: H8, H9, H11, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Detailed description of necessary equipment and the refluxing process or suitable labelled diagram - Justification given for equipment, chemicals (including catalyst) and refluxing - Specifies chemicals for making the ester including the catalyst - Answer illustrated with correct equation.	5–6
- Describes equipment and refluxing process or draws labelled diagram - Gives some reasoning for equipment and method used - Identifies some chemicals used	3–4
Outlines some equipment or chemicals used in esterification in general	1–2

Question 23 (a) (1 mark)

Outcomes assessed: H1, H6

MARKING GUIDELINES

Criteria	Marks
Correct definition stated	1

Question 23 (b) (3 marks)

Outcomes assessed: H6, H10, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the molar mass (units required)	3
- States the correct moles of HCl used and hence moles of NaX in titration and in 100 mL sample OR - Correct process used throughout, but error in calculation or transcription	2
- Balanced chemical equation for the reaction OR - States correct moles HCl used OR - Indicates moles in 100 mL is 5x moles in 20 mL sample OR - Provides a definition of molar mass OR - Gives correct answer and shows no working (units not required)	1

Question 24 (a) (1 mark)

Outcomes assessed: H3

MARKING GUIDELINES

Criteria	Marks
Any relevant use stated	1

Question 24 (b) (2 marks)

Outcomes assessed: H8, H13

MARKING GUIDELINES

Criteria	Marks
- States that the yield is increased - Explains the increase in terms of Le Chatelier's principle and the removal of product	2
- States that the yield increased OR - Indicates that equilibrium shifts right	1

Question 24 (c) (3 marks)

Outcomes assessed: H4, H8, H10

MARKING GUIDELINES

Criteria	Marks
- Indicates the conflicting effect of temperature on reaction rate and yield - Explains the effect of pressure on yield - Identifies that a compromise set of conditions must be used in equilibrium reactions	3
- Identifies that reaction has optimum conditions of temperature and pressure that affect reaction rate OR yield OR safety conditions - Explains how temperature OR pressure affect yield OR reaction rate	2
Identifies that reaction has optimum conditions of temperature and pressure that affect reaction rate OR yield OR safety conditions	1

Question 25 (6 marks)

Outcomes assessed: H4, H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
- Displays extensive knowledge of possible different reaction products, either due to identified varying reaction conditions or due to the presence of impurities using an appropriate example - Explains the need for monitoring these products to avoid problems	5–6
- Identifies at least one variation in reaction products using an appropriate example AND - Relates formation of this product to the need for monitoring to avoid problems	3–4
- Describes a specific chemical reaction OR - States purpose of monitoring	1–2

Question 26 (a) (1 mark)*Outcomes assessed: H13***MARKING GUIDELINES**

Criteria	Marks
- All points plotted correctly on grid provided - And line of best fit or points joined	1

Question 26 (b) (1 mark)*Outcomes assessed: H12, H14***MARKING GUIDELINES**

Criteria	Marks
Gives a value between 1.4 and 1.6	1

Question 26 (c) (2 marks)*Outcomes assessed: H4, H14***MARKING GUIDELINES**

Criteria	Marks
A hypothesis which could be tested and is logical based on the data	2
Any inference based on the patterns shown in the table	1

Question 27 (4 marks)

Outcomes assessed: H4, H5, H13, H14

MARKING GUIDELINES

Criteria	Marks
Compares the environmental effects of ozone in the upper and lower atmosphere, relating these to concentrations	4
Compares the environmental effects of ozone in the upper and lower atmosphere	3
- Identifies the concentration of ozone (in the upper atmosphere) and one related environmental effect OR - Identifies the concentration of ozone (in the lower atmosphere) and one related environmental effect OR - Identifies two environmental effects of ozone	2
- Identifies that the upper atmosphere has a higher concentration of ozone than the lower atmosphere OR - Identifies an environmental effect of ozone	1

Question 28 (a) (i) (1 mark)

Outcomes assessed: H7, H13

MARKING GUIDELINES

Criteria	Marks
Any correct definition of electrolysis	1

NB. do not accept an example isolated from a definition.

Question 28 (a) (ii) (2 marks)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
Correctly identifies that both electrolyses produce chlorine gas but that the other product for each is different and names that product	2
- Correctly identifies one product for both reactions OR - Correctly identifies two products from one reaction	1

Question 28 (b) (i) (2 marks)*Outcomes assessed: H10***MARKING GUIDELINES**

Criteria	Marks
• Writes the correct balanced equation using chemical symbols	2
• Writes a correct word equation OR • Gives the correct formula for carbon monoxide and for chlorine gas	1

Question 28 (b) (ii) (2 marks)*Outcomes assessed: H3, H8***MARKING GUIDELINES**

Criteria	Marks
• Correctly identifies one factor which influences the equilibrium and describes the way in which this factor can be used to maximise production of phosgene, using their answer in part (i)	2
• Correctly identifies one factor which influences the equilibrium OR • Relates Le Chatelier's principle to maximising production by indicating correct equilibrium shift	1

Question 28 (c) (5 marks)

Outcomes assessed: H3, H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
- Explains the importance of sulfuric acid as an industrial chemical based on a number of its properties and related use - Includes balanced chemical equations related to at least two properties	5
Relates at least one use of sulfuric acid to its properties and includes a balanced chemical equation related to one property	4
Relates one use of sulfuric acid to a property and includes a balanced equation for that property	3
- States one property and one use of sulfuric acid OR - States two properties of sulfuric acid OR - States two uses of sulfuric acid	2
- States one property of sulfuric acid OR - States one use of sulfuric acid OR - Correct relevant equation	1

Question 28 (d) (i) (1 mark)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Saponification	1

Question 28 (d) (ii) (2 marks)*Outcomes assessed: H12***MARKING GUIDELINES**

Criteria	Marks
Identifies suitable equipment and identifies both reactants and the heating process	2
- Identifies both reactants OR - Identifies one reactant and correct process	1

Question 28 (d) (iii) (3 marks)*Outcomes assessed: H12***MARKING GUIDELINES**

Criteria	Marks
Describes a safety risk relevant to the procedure outlined and suggests a safe working practice to minimise the risk	3
States one hazard and one safe work practice	2
States one hazard or one safe work practice	1

Question 28 (e) (7 marks)

Outcomes assessed: H4, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of the Solvay process supported by at least one relevant chemical relationship • Describes environmental issues or potential issues associated with the process • Makes a judgement about the value of the way in which the environmental issues are addressed	6–7
• Demonstrates a knowledge of the Solvay process • Identifies environmental issues or potential environmental issues associated with either reactants, intermediates or products of the process • Demonstrates knowledge of procedures used to address environmental issues	4–5
• Identifies some reactants and/or products of the Solvay process OR • Some knowledge of procedures used to control environmental issues or potential environmental issues OR • States some environmental issues associated with the process	2–3
• Identifies a reactant or product of the Solvay process OR • States a procedure used to control a potential environmental problem associated with the process OR • States an environmental problem associated with the process	1

Question 29 (a) (i) (1 mark)*Outcomes assessed: H6***MARKING GUIDELINES**

Criteria	Marks
• Iron or steel	1

Question 29 (a) (ii) (2 marks)*Outcomes assessed: H6, H7***MARKING GUIDELINES**

Criteria	Marks
• Correctly identifies formation of an oxide which creates an impervious layer to oxygen	2
• Correctly identifies the formation of an oxide layer OR • Correctly identifies the formation of an impervious layer OR • Correctly identifies that aluminium is a passivating metal	1

Question 29 (b) (i) (1 mark)*Outcomes assessed: H8***MARKING GUIDELINES**

Criteria	Marks
• Correctly identifies a metal	1

Question 29 (b) (ii) (3 marks)***Outcomes assessed: H4, H8*****MARKING GUIDELINES**

Criteria	Marks
- Identifies that corrosion is prevented by preferential oxidation of the sacrificial anode AND - Relates this to the relevant reduction potentials for an appropriate sacrificial anode, and iron	3
- Identifies that corrosion is prevented by preferential oxidation of the sacrificial anode OR - States relevant reduction potentials for an appropriate sacrificial anode, and iron	2
Identifies that corrosion of the hull metal is prevented	1

Question 29 (c) (5 marks)

Outcomes assessed: H3, H8, H13

MARKING GUIDELINES

Criteria	Marks
Identifies carbon as an essential component of steel and demonstrates a detailed knowledge of the effect of addition of two or more elements (one of which may be carbon) to iron on the properties and subsequent use of steel	5
- Describes 2 or more examples of how the addition of different elements changes the properties of iron and determines the subsequent use of the steel OR - Describes how the addition of different proportions of an element to iron can produce varying properties and different uses of steels OR - Provides a detailed description of how the added element alters the properties of the steel	4
- Describes how the addition of at least one other element changes the properties of iron and determines the subsequent use of the steel OR - Gives 2 or more examples of how the addition of an element changes the properties of iron OR - Gives 2 or more examples of how the addition of an element determines the subsequent uses of iron	3
- Identifies 2 or more elements that are added to iron to form a steel OR - Identifies how the addition of an element changes the properties of iron OR - Identifies that the addition of an element to iron determines the subsequent use of the steel	2
- Identifies one element that is added to iron to form a steel OR - Identifies one property of iron that can be enhanced OR - States one use of a steel	1

Question 29 (d) (i) (1 mark)

Outcomes assessed: H8, H13

MARKING GUIDELINES

Criteria	Marks
Any correct definition of corrosion	1

Question 29 (d) (ii) (2 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
- Identifies a suitable corrosive medium - Identifies suitable equipment - Describe how the rate of corrosion will be monitored	2
Identifies a suitable corrosive medium and identifies suitable equipment	1

Question 29 (d) (iii) (3 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Clearly distinguishes between accuracy and reliability with examples of how to improve both	3
- States one way to improve accuracy and one way to improve reliability OR - States more than one way to improve accuracy OR - States more than one way to improve reliability OR - Distinguishes between accuracy and reliability in general terms	2
States one way to improve accuracy or one way to improve reliability	1

Question 29 (e) (7 marks)

Outcomes assessed: H4, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Demonstrates an extensive knowledge and correct order of the steps in all 3 processes for a named metal AND - Makes a judgement on the relevance of each step in the recovery of the artefact	6–7
- Demonstrates a thorough knowledge of a number of steps in all 3 processes OR - Describes two processes with an evaluation of one process.	4–5
- Outlines cleaning or preserving or stabilising, and describes at least one step in each of the other two processes OR - Gives examples of isolated steps across cleaning, preserving and stabilising	2–3
States one step in cleaning or preserving or stabilising.	1

Question 30 (a) (i) (1 mark)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Adenosine triphosphate	1

Question 30 (a) (ii) (2 marks)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies the biologically important part AND - Explains the reaction that releases energy OR - An equation outlining the process (not including water)	2
- Correctly identifies the biologically important part OR - An equation outlining the process	1

Question 30 (b) (i) (2 marks)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Identifies the molecule and states its function	2
- Identifies the molecule OR - States its function	1

Question 30 (b) (ii) (2 marks)

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
Relates oxidative phosphorylation to ATP production and identifies the link to the oxidation of NADH or FADH₂	2
Relates oxidative phosphorylation to ATP production or to the oxidation of NADH or FADH₂	1

Question 30 (c) (5 marks)

Outcomes assessed: H2, H9, H13

MARKING GUIDELINES

Criteria	Marks
• Relates properties of enzyme specificity and catalysis to how the model works	4–5
• States one property of an enzyme and describes/draws a model of enzyme activity	2–3
• Describes/draws a model of enzyme activity	1

Question 30 (d) (i) (1 mark)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
• Defines viscosity	1

Question 30 (d) (ii) (2 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Describes an appropriate procedure	2
• States a time comparison is necessary OR • States identical experimental conditions	1

Question 30 (d) (iii) (3 marks)***Outcomes assessed: H12*****MARKING GUIDELINES**

Criteria	Marks
Clearly distinguishes between accuracy and reliability with examples of how to improve both	3
- States one way to improve accuracy and one way to improve reliability OR - States more than one way to improve accuracy OR - States more than one way to improve reliability OR - Distinguishes between accuracy and reliability in general terms	2
States one way to improve accuracy or one way to improve reliability	1

Question 30 (e) (7 marks)

Outcomes assessed: H7, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Relates type 1 and type 2 skeletal muscle to the correct respiratory pathways and rate and amount of ATP production by both aerobic and anaerobic pathways during different types of exercise AND - Shows how these respiratory pathways are interrelated. Equations or a flow chart are used to summarise the respiratory pathways	6–7
- Summarises both aerobic and anaerobic respiratory pathways and interrelates these in terms of their energy production. Molecules are correctly identified OR - Identifies both types of muscle and describes the energy requirement for different types of exercise and connects these to the correct respiratory pathways and amount of ATP produced without necessarily showing the interrelationship of the pathways	4–5
- Relates the type of muscle to the correct respiratory pathway and amount of ATP OR - Identifies all reactants and products including the number of ATP either anaerobic or aerobic respiration OR - Compares energy production in both aerobic and anaerobic respiration OR - Compares the type of muscle used and the energy requirement for different type of exercise	2–3
- Identifies a muscle type OR - Mentions a condition for energy production in muscle OR - Identifies a reactant or product of glucose metabolism	1

Question 31 (a) (i) (1 mark)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Any suitable mineral source of a pigment	1

Question 31 (a) (ii) (2 marks)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Describes how a separation process can be used to obtain a pigment	2
Partially correct description of a separation process	1

Question 31 (b) (i) (2 marks)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
Identifies barium and chromium only	2
1 correct element identified OR	1
Identifies 2 correct elements and 1 other that is incorrect	

Question 31 (b) (ii) (2 marks)

Outcomes assessed: H2, H6

MARKING GUIDELINES

Criteria	Marks
Explains that an element must be excited for a subsequent emission to occur and that only certain energies are emitted from an element and that these correspond to certain frequencies/wavelength of light	2
States that an element must be excited to a higher energy state OR	1
States that only certain energies are emitted from an element	

Question 31 (c) (5 marks)

Outcomes assessed: H2, H6, H13

MARKING GUIDELINES

Criteria	Marks
- Provides a detailed description of the Bohr model of the atom - Identifies a merit of the model and a limitation of the model	5
- Describes the Bohr model of the atom and identifies a merit of the model OR - Describes the Bohr model and identifies a limitation	3–4
Partially describes the Bohr model of the atom	2
States the Bohr model refers to electrons	1

Question 31 (d) (i) (1 mark)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Any transition element given	1

Question 31 (d) (ii) (2 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Nominates reactants and colour change	2
- Nominates reactants OR - Correctly links a colour to a metal's oxidation state	1

Question 31 (d) (iii) (3 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Describes a safety risk and suggest a safe working practice to minimise the risk	3
States one hazard and one safe work practice	2
- States one hazard OR - One safe work practice	1

Question 31 (e) (7 marks)

Outcomes assessed: H6, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Discusses the composition of 2 pigments including the formula of each coloured compound, and the oxidation state of each metal involved AND - Relates the colour of each pigment to electronic transitions eg between d orbitals in transition metals	6–7
- Discusses the composition of 2 pigments including the formula of each coloured compound, and the oxidation state of each metal involved OR - Relates the colour of each pigment to electron transition eg between d orbitals in the two transition metals	4–5
Names two pigments and gives the chemical composition of one of the pigments	3
- Names two pigments OR - Names two transition metals OR - Names a pigment and gives its chemical composition	2
- Names a pigment OR - Names a transition metal	1

Question 32 (a) (i) (1 mark)*Outcomes assessed: H9, H13***MARKING GUIDELINES**

Criteria	Marks
• A correct definition is given	1

Question 32 (a) (ii) (2 marks)*Outcomes assessed: H9***MARKING GUIDELINES**

Criteria	Marks
• Correctly identifies a class of organic compounds AND • Describes a correct test with correct results to identify that class	2
• Correctly identifies a class of organic compounds OR • States a test that identifies a class of organic compounds	1

Question 32 (b) (i) (1 mark)*Outcomes assessed: H14***MARKING GUIDELINES**

Criteria	Marks
• Tallow	1

Question 32 (b) (ii) (3 marks)

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
Relates the fatty acid structure and length of hydrocarbon chain in fatty acids and the carboxyl group to the solubility of fatty acids in water	3
- Describes the fatty acid molecule and describes solubility in water of the carboxylic acid group OR - Relates the length of hydrocarbon chain in fatty acids to the solubility of fatty acids in water	2
- Describes the fatty acid molecule OR - Describes solubility in water of the carboxylic acid group	1

Question 32 (c) (5 marks)

Outcomes assessed: H2, H3, H6

MARKING GUIDELINES

Criteria	Marks
Assess the usefulness of mass spectrometry in forensic chemistry in terms of a number of its advantages such as determining RMM, range of elemental isotopes present, relative amounts of isotopes, very small samples	4–5
Describes the usefulness of mass spectrometry in forensic chemistry with a suitable example	2–3
Identifies one area of usefulness of mass spectrometry e.g. very small samples or range of elemental isotopes present	1

Question 32 (d) (i) (1 mark)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
A technique correctly identified	1

Question 32 (d) (ii) (2 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Describes the method and details of the materials used	2
• Gives details of the materials used	1

Question 32 (d) (iii) (3 marks)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Clearly distinguishes between accuracy and reliability with examples of how to improve both	3
• States one way to improve accuracy and one way to improve reliability OR • States more than one way to improve accuracy OR • States more than one way to improve reliability	2
• States one way to improve accuracy or one way to improve reliability OR • Distinguishes between accuracy and reliability in general terms	1

Question 32 (e) (7 marks)

Outcomes assessed: H1, H3, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Relates the unique nature of individual DNA to the use of DNA analysis in forensic chemistry - Discusses the role of DNA analysis in forensic chemistry by comparing it to another method and referring to points for and against the use of the technique	6–7
- Relates the unique nature of individual DNA to the use of DNA analysis in forensic chemistry AND - Compares the role of DNA analysis in forensic chemistry with another method	4–5
Identifies the unique nature of individual DNA and describes at least one use of DNA analysis in forensic chemistry	2–3
- Identifies the unique nature of individual DNA OR - Identifies a use of DNA analysis in forensic chemistry	1